

Propagator Eval – Reconciliation Table

Google Doc vs. local protocol files vs. execution-ready plan

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Sources compared.

1. Google Doc: EXPERIMENT_PROTOCOL.md (Drive doc id 13dusgXpzxQnYrDnS2JphJ-aBdwjjzaiWeqvXMjMMkck)
2. Local protocol: /home/bosho/FP/experiments/propagator_eval/V2.md
3. Local execution docs: EXPERIMENT_OBJECTIVES.md, RESULTS_AND_IMPLEMENTATION.md, RUNBOOK.md, and the YAML configs in configs/

Executive summary

- The **core design is consistent** across the Google Doc and local files.
- The **local plan is more execution-ready** because it adds concrete configs, runners, schemas, and post-processing steps.
- There are **three real reconciliations** to lock down before launch:
 1. include or exclude the heterophilic downstream block,
 2. decide whether boundary_coverage is required and spectral_fidelity is deferred,
 3. use the run counts from RUNBOOK.md carefully because it rolls ablations into the intrinsic total.
- Recommended outcome: use the **local execution plan** as the authoritative run plan, with one explicit note that boundary_coverage is in-scope and spectral_fidelity is not required for the first run.

Reconciliation table

Topic	Google Doc	Local files	Recommended final plan
Paper framing	Intrinsic evaluation of communication-free propagation; no need to beat generator/pre-communication families on accuracy.	Same framing, but local docs spell it out more explicitly.	No action needed. Keep the stronger local wording when briefing others.
Primary operators	Adj, Diffusion, Chebyshev, APPNP, Random Walk; Heat Kernel exact as reference.	Same operator set, with code names and config hooks already present.	Treat as fully aligned.
Datasets	Cora, Citeseer, Pubmed, OGBN-Arxiv, Texas, Wisconsin.	Same six datasets; configs already reflect this.	Treat as fully aligned.
Intrinsic matrix	270 primary + 2 heat-kernel + 45 hop-2 = 317 core runs. Ablations listed separately.	Same core counts. RUNBOOK.md reports 359 because it includes 42 ablation runs in the L1-L3 total.	Use 317 as the core intrinsic matrix; cite 359 only when explicitly counting ablations.
Downstream homophilic matrix	400 operator runs + 80 zero-hop + 60 oracle = 540.	Same 540 homophilic runs.	Treat as fully aligned.
Downstream heterophilic block	Not included in the original downstream total.	Added explicitly in V2.md, EXPERIMENT_OBJECTIVES.md, RUNBOOK.md, and L4_downstream_heterophilic_block.yaml. 40 runs on Texas/Wisconsin with Adj, Diff, APPNP, and zero-hop.	Adopt it. It is a small but meaningful theory-validation exercise and is already configured.
Total Layer 4 count	540 runs.	580 runs because of the extra heterophilic block.	Use 580 as the final execution total.
Intrinsic metrics	Calls for MSE, cosine similarity, recovery ratio, and spectral_fidelity.	Implemented docs and code emphasize MSE, cosine, recovery ratio, and boundary_coverage.	Make boundary_coverage required. Defer spectral_fidelity unless there is a concrete analysis need later.
Output schema	Broad schemas described, but fewer operational details.	Precise JSON layout, folder layout, and post-hoc gap computation are defined.	Use the local schemas as authoritative for produced artifacts.

Post-processing	Gap-closed metric described conceptually.	Explicit --compute-gaps workflow after baselines complete.	Use the local workflow.
Execution readiness	Protocol-level specification.	Execution-ready: configs exist, runners exist, modes are wired, and output paths are specified.	Use local files as the operational source of truth.

Run-count reconciliation

Category	Google Doc	Local plan	Reconciled interpretation
Layer 1-3 core	317	317	Aligned.
Ablations	42	42	Aligned.
Layer 1-3 if ablations included	Not rolled up	359	Local RUNBOOK.md bundles ablations into the intrinsic total.
Layer 4 homophilic	540	540	Aligned.
Layer 4 heterophilic extension	0	40	Intentional local extension.
Layer 4 total	540	580	Use 580.
Grand total with ablations	899	939	Difference is exactly the extra 40 heterophilic downstream runs.

Recommended final execution plan

Authoritative operational plan

- Use the local execution stack (RUNBOOK.md + configs/ + existing runners) as the source of truth for actual runs.
- Keep the Google Doc as the high-level conceptual protocol.
- Explicitly annotate one update when sharing with collaborators: the local plan adds a 40-run heterophilic downstream validation block, raising Layer 4 from **540** to **580** and the grand total from **899** to **939**.
- Lock the intrinsic metric set for the first pass as: mse, cosine_sim, recovery_ratio, boundary_coverage, iteration count, residuals, convergence flag, and wall-clock time.
- Treat spectral_fidelity as **deferred / optional**, not launch-blocking.

Practical note

The repo is already close to execution-ready: the intrinsic runner, downstream runner, intrinsic config, and the new propagation modes are present in /home/bosho/FP. This document therefore reconciles **protocol differences**, not fundamental implementation blockers.